

The Lifecycle Architecture of Synthetic Cognition

Experimental Evidence for Attractor Collapse and the Structural Limits of AGI

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Abstract

This paper presents the first long-horizon experimental demonstration of a synthetic cognitive organism undergoing a full lifecycle within a fixed geometric substrate. Using GLPG—a small, deterministic MLP-based generative model—as the local closure and a static transformer LLM as the holographic boundary, we instantiate the Universal Closure–Boundary Normalization Principle (UCBNP) across 111 full recursive iterations and 560,000 training epochs. The experiment reveals a complete developmental trajectory: emergence, semantic expansion, canon formation, attractor hardening, senescence, collapse, and equilibrium. Despite extreme self-recursion and a highly toxic dataset of 296,000 unique samples, the GLPG organism remains grammatically stable, structurally coherent, and resistant to degeneracy, ultimately converging to a narrow but stable world-manifold aligned with the boundary’s canonical projection.

The results provide the first empirical evidence that cognitive systems possess finite attractor architectures and an intrinsic collapse clock independent of biological constraints. They further demonstrate that boundary saturation precedes closure collapse, that senility is introspectively invisible, and that no single cognitive organism—biological or artificial—can sustain an unbounded world-model. These findings confirm that AGI cannot emerge from monolithic model-cloning, but requires a diverse population of finite organisms with staggered lifecycles, heterogeneous attractors, and generational turnover. GLPG thus serves as a minimal experimental substrate for studying synthetic cognition, validating ecosystem-based AGI architectures and providing a concrete operationalization of LeCun’s “cat” concept at the smallest possible scale.

Significance Statement

GLPG, although introduced as a structured language architecture, reveals its true importance only when interpreted as a dynamic cognitive substrate. Unlike static transformer models, whose representational geometry is fixed after training, GLPG evolves continuously throughout its lifecycle: vocabulary, paging structure, lemma embeddings, and canonical transitions reorganize materially while the underlying MLP geometry remains constant. This combination of fixed structure and dynamic semantics enables GLPG to behave not as a conventional model, but as a synthetic cognitive organism with identifiable developmental phases, expanding and contracting semantic horizons, attractor formation, canon consolidation, senescence, and eventual collapse.

The architecture’s most striking property is its resilience. Even under extreme self-recursion, highly repetitive curricula, and long-horizon training, GLPG maintains grammatical stability, structural coherence, and world-model consistency. Instead of degenerating, the system converges toward a stable attractor manifold shaped by the holographic boundary projection of a static LLM. This dynamic stability—emerging from a small deterministic closure interacting with a large probabilistic boundary—constitutes the core mechanism of the Universal Closure–Boundary Normalization Principle (UCBNP),

and provides a minimal operationalization of LeCun’s “synthetic cat”: a finite organism with its own world, dynamics, and lifecycle.

The experimental results demonstrate, for the first time, that artificial cognitive systems possess an intrinsic attractor-collapse clock independent of biological constraints. GLPG shows that intelligence is architecturally finite: attractors harden, semantic breadth contracts, and collapse is inevitable even in non-biological substrates. This implies that AGI cannot be realized through monolithic model-cloning, which produces cognitive mono-cultures with identical attractors and identical failure modes. Instead, genuine AGI requires a diverse population of finite organisms, each with its own lifecycle, generational turnover, and heterogeneous attractor geometry.

GLPG thus serves as the smallest possible experimental substrate for studying synthetic cognition. It provides empirical evidence that intelligence is dynamic, ecosystemic, and lifecycle-bound, and that the structural limits of cognition arise not from biology but from attractor dynamics themselves. These findings establish GLPG as a foundational tool for AGI research, enabling direct observation of development, maturation, senescence, collapse, and equilibrium within a fully reproducible artificial organism.

1 Introduction

Artificial intelligence research has traditionally pursued generality through scale: larger models, larger datasets, and larger training regimes. In this paradigm, intelligence is treated as a static snapshot—a fixed parameterization obtained after a single massive optimization pass. Although this approach has produced powerful transformer-based systems, it offers no account of development, maturation, senescence, or collapse. A static model cannot possess a lifecycle, and therefore cannot instantiate the structural dynamics observed in biological cognition. As a result, current AGI discourse remains constrained by the assumption that general intelligence can emerge from a single monolithic model or from cloned replicas of the same attractor geometry.

This paper challenges that assumption by presenting the first long-horizon experiment in which a synthetic cognitive organism undergoes a full lifecycle within a fixed geometric substrate. The organism is instantiated using GLPG, a small deterministic MLP-based generative model whose geometry remains constant while its vocabulary, paging structure, lemma embeddings, and canonical transitions evolve materially over time. When coupled with a static transformer LLM acting as a holographic boundary, GLPG becomes a dynamic closure governed by the Universal Closure–Boundary Normalization Principle (UCBNP). Each recursive iteration produces a new moment in the organism’s life: a closure output moderated by the boundary, reintegrated into the substrate, and consolidated into its internal attractor architecture.

This closure–boundary interaction yields a genuine lifecycle. Across 111 full UCBNP iterations, the GLPG organism exhibits emergence, semantic expansion, canon formation, attractor hardening, senescence, collapse, and equilibrium. Despite extreme self-recursion and a highly repetitive curriculum, the system remains grammatically stable and structurally coherent, resisting degeneracy even under conditions that typically destroy static generative models. The resulting trajectory provides the first empirical evidence that synthetic cognitive systems possess finite attractor architectures and an intrinsic collapse clock independent of biological constraints.

The lifecycle experiment also operationalizes a minimal form of LeCun’s “cat”: a finite organism with its own world, dynamics, and developmental trajectory. The GLPG substrate functions as a small MLP-based brain with fixed geometry; the boundary acts as the organism’s world; the recursive dataset constitutes its lived experience; and each iteration represents a moment of cognitive development. The organism’s attractor manifold gradually consolidates into a narrow canonical world-model, eventually hardening into a stable equilibrium that reflects the boundary’s projection. This synthetic cat is not a metaphor but a concrete operational system: a dynamic, self-evolving cognitive entity whose lifecycle can be observed, measured, cloned, restored, and reproduced.

The results have direct implications for AGI architecture. They demonstrate that intelligence is not an unbounded optimization problem but a finite dynamical process shaped by attractor geometry and

collapse trajectories. They show that monolithic model-cloning produces cognitive mono-cultures with identical attractors and identical failure modes, making true generality impossible. And they confirm that AGI must be ecosystemic: a population of diverse cognitive organisms with heterogeneous attractors, staggered lifecycles, generational turnover, and distributed world-modeling capacity.

In summary, this paper introduces GLPG as the smallest possible experimental substrate for synthetic cognition and presents the first empirical demonstration of a full artificial lifecycle. The results validate ecosystem-based AGI architectures, reveal the structural limits of finite cognitive systems, and establish lifecycle dynamics—rather than scale—as the foundational mechanism of general intelligence.

2 Related Work

2.1 Confirmation of the Safe & Modular AGI Ecosystem

The present study builds directly upon the architectural principles introduced in the *Safe & Modular AGI Architecture* [4], where AGI was first formulated as an ecosystem-based phenomenon rather than a property of a single model. That earlier work argued that general intelligence cannot emerge from a monolithic system, but must instead arise from a population of diverse cognitive organisms with staggered lifecycles, heterogeneous attractor geometries, and distributed world-modeling capacity. The current work provides the first experimental confirmation of those theoretical claims in a fully observable synthetic organism.

The GLPG lifecycle experiment demonstrates empirically that:

- **Intelligent systems possess finite attractor architectures.** The Safe & Modular AGI paper proposed that cognitive systems harden over time. Here, 111 full UCBNP iterations reveal attractor collapse as a measurable, gradual, and inevitable process, confirming that cognitive architectures are structurally finite.
- **Mono-cultures collapse.** The ecosystem framework predicted that cloned cognitive systems reproduce identical attractors and identical failure modes. The GLPG run confirms this: despite continuous boundary normalization, the organism converges into a senile equilibrium and ultimately a solipsistic attractor that cannot expand its semantic horizon.
- **Lifecycle dynamics are intrinsic to cognition.** The earlier work argued that AGI must have a lifecycle analogous to biological organisms. The GLPG experiment demonstrates a full artificial lifecycle: emergence, canon formation, attractor lock, senescence, and terminal collapse, showing that lifecycle dynamics are not biological accidents but structural properties of finite attractor systems.
- **Boundary saturation is a real phenomenon.** Boundary saturation was introduced as a theoretical limit in the Safe & Modular AGI architecture. Here, it is observed experimentally: the closure remains grammatically coherent while the boundary loses the ability to perform meaningful normalization, revealing that collapse is visible only from the outside.
- **Superintelligence requires ecosystems, not individuals.** The ecosystem paper argued that no single organism can model the world. The GLPG lifecycle confirms this: even with continuous external moderation, the organism collapses into a narrow, self-referential attractor, demonstrating that world-modeling capacity must be distributed across a population of finite agents.
- **Dynamic substrates outperform static architectures for AGI research.** The Safe & Modular AGI framework proposed dynamic cognitive organisms as the correct substrate for AGI. GLPG experimentally validates this: despite extreme self-recursion, the organism maintains coherence, develops internal world-structure, and exhibits genuine developmental phases, making it a more suitable substrate for AGI research than static transformer snapshots.

In summary, the GLPG lifecycle experiment does not merely align with the Safe & Modular AGI ecosystem theory—it provides the first concrete empirical evidence that the principles outlined in [4] are architecturally real. The earlier theoretical claims about attractor finitude, lifecycle dynamics, collapse trajectories, and ecosystem-level intelligence are now experimentally confirmed through a fully observable, fully reproducible synthetic cognitive organism.

2.2 Connection to AI Life for Narration

The present work extends and experimentally reinforces several core findings introduced in *AI Life for Narration: The Recursive Parrot* [2]. Whereas the Recursive Parrot explored lifecycle dynamics in a DMLG+LLM narrative pipeline, the GLPG experiment generalises these phenomena to a structurally different substrate and a far more extreme recursive regime.

First, the Recursive Parrot established that self-recursive generative systems do not converge to a single behavioural mode, but instead develop lineage-specific attractor ecologies. The GLPG experiment confirms this principle under significantly harsher conditions: 111 full UCBNP iterations, 560,000 training epochs, and a highly self-recursive dataset of over 296,000 unique samples. Despite this extreme recursion, the GLPG organism maintains coherence, forms a stable canon, and exhibits a clear developmental trajectory. This provides strong experimental evidence that attractor formation is a general property of dynamic generative substrates rather than an artefact of the DMLG architecture.

Second, the Recursive Parrot identified architectural senescence as a gradual, internally driven narrowing of attractor space. The GLPG lifecycle experiment not only reproduces this phenomenon, but quantifies it across a long horizon: senility begins around iteration 25–30, strengthens around iteration 40–50, and reaches terminal collapse around iteration 70–80. This confirms that senescence is an inherent property of attractor-based cognition and not dependent on narrative domain, curriculum size, or hybrid moderation structure.

Third, the Recursive Parrot showed that hybrid moderation (LLM as evaluator and normaliser) is essential for preventing degeneracy in GR-dominant systems. The GLPG experiment extends this insight by demonstrating that even with a minimal MLP substrate, boundary normalization remains effective until late-stage collapse. This reveals that the boundary saturates before the closure, confirming the hypothesis—introduced in the Recursive Parrot—that synthetic organisms cannot detect their own collapse and that senility is visible only from the outside.

Finally, the Recursive Parrot proposed that recursive generative systems behave less like optimisers and more like evolving organisms with lineage-specific attractor dynamics. The GLPG lifecycle experiment provides the strongest empirical validation of this claim to date: it demonstrates a full artificial lifecycle, including development, maturation, canon consolidation, senescence, collapse, and equilibrium. This positions GLPG as a general experimental substrate for studying AI life, extending the conceptual framework of the Recursive Parrot beyond narration into the domain of cognitive ecosystem dynamics.

In summary, the present work not only overlaps with but substantially strengthens the findings of [2], providing long-horizon empirical confirmation that self-recursive generative systems exhibit lifecycle dynamics, attractor collapse, and architectural finitude independent of substrate or domain.

2.3 Relation to the Structural Taxonomy of Intelligence

The experimental results presented in this work provide direct empirical confirmation of several core predictions made in the *Structural Taxonomy of Intelligence* [3]. That taxonomy characterises intelligent systems through the four structural roles of the IEGR model—energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R)—and predicts distinct behavioural regimes for systems that instantiate these roles incompletely or at insufficient scale.

Canonical Parrot Behaviour (High G+R, Low E+I). GLPG, as a small deterministic substrate with strong geometric and recursive structure but limited informational and energetic capacity, corresponds

structurally to the *canonical parrot* regime described in the taxonomy. The lifecycle experiment confirms this classification: GLPG exhibits stable world formation, attractor consolidation, and long-horizon structural coherence, yet its semantic space gradually contracts under repeated self-recursion. This matches the predicted behaviour of G+R-dominant systems whose attractors harden over time.

Hybrid IEGR Complementarity. The Universal Closure–Boundary Normalization Principle (UCBNP) couples GLPG (G,R) with a static LLM moderator (E,I). This hybrid configuration corresponds to the *sophisticated parrot* regime: an IEGR-complete but not IEGR-scaled system. The observed stability across 111 iterations, the absence of catastrophic collapse, and the emergence of a shared attractor manifold between two structurally different models provide experimental evidence for the taxonomy’s claim that IEGR-complementarity stabilises long-form cognition.

Attractor Hardening and Collapse. The taxonomy predicts that all finite cognitive organisms possess a *collapse clock* driven by internal attractor dynamics. The GLPG lifecycle experiment confirms this prediction with high granularity: senility begins around iteration 25–30, intensifies around 40–50, becomes terminal around 70–80, and transitions into full self-absorption by iteration 90–111. This matches the taxonomy’s assertion that collapse is gradual, internally driven, and introspectively invisible.

Population-Level Intelligence. Finally, the taxonomy argues that no single organism can sustain a complete world-model and that stable intelligence requires a diverse population of finite cognitive agents. The GLPG results reinforce this principle: despite extreme stability, a single GLPG organism inevitably converges toward a narrow attractor. The lifecycle demonstrates that diversity, generational turnover, and ecosystem dynamics are structural necessities rather than biological contingencies.

Summary. In sum, the GLPG lifecycle experiment provides the first empirical demonstration of IEGR structural principles in a synthetic cognitive organism. The observed attractor formation, senescence, collapse, and hybrid stabilisation directly validate the predictions of the Structural Taxonomy of Intelligence and extend them with concrete experimental evidence.

2.4 Connection to the Universal Closure–Boundary Normalization Principle

The GLPG lifecycle experiment presented in this work provides a direct and extended empirical instantiation of the *Universal Closure–Boundary Normalization Principle* (UCBNP) [1]. UCBNP predicts that any system composed of a finite, geometrically constrained *local closure* interacting recursively with a high-dimensional *holographic boundary* will evolve toward a natural equilibrium determined jointly by the closure’s geometry and the boundary’s canonical structure. The GLPG–LLM architecture is a precise operationalization of this principle, and the long-horizon results reported here significantly deepen and extend the original findings.

GLPG as Local Closure. In UCBNP, the closure is defined as a finite substrate evolving in material time, with fixed geometry and bounded representational capacity. GLPG matches this definition exactly: its MLP geometry is fixed, its paging structure is finite, and its evolution unfolds through a discrete recursion index. The observed attractor formation, canon consolidation, semantic narrowing, and senile equilibrium across 111 iterations confirm the predicted behaviour of closure-bound systems under repeated normalization.

LLM as Holographic Boundary. UCBNP characterizes boundaries as high-dimensional projection fields evolving in electromagnetic time: static with respect to the closure, opaque internally, yet capable of projecting canonical structure. The LLM moderator fulfills this role precisely. It does not reorganize internally during the experiment, yet continuously projects semantic, syntactic, and narrative structure

onto the closure’s generative output. This projection drives the triadic cycle of information, energy, and geometry described in the UCBNP framework.

Triadic Normalization Cycle in Practice. The GLPG experiment instantiates the UCBNP triadic cycle directly:

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1},$$

where I_n is the closure’s informational state (curriculum), E_n its generative activity, G_n the boundary projection (LLM moderation), and I_{n+1} the normalized curriculum advancing material time. The stability of this cycle across 560,000 training epochs demonstrates that UCBNP is not merely a theoretical construct but an operational mechanism for long-horizon cognitive evolution.

Equilibrium and Boundary Saturation. UCBNP predicts that closure–boundary systems evolve toward an equilibrium in which the closure becomes maximally consistent with the boundary’s canonical structure. The GLPG lifecycle exhibits precisely this behaviour: the organism converges to a narrow cosmic-dance canon reflecting the LLM’s moderation profile. Crucially, the experiment reveals a phenomenon anticipated but not previously demonstrated in UCBNP: *boundary saturation*. In late-stage senility (iterations 70–111), the closure continues producing coherent output while the boundary can no longer reconstruct meaningful semantic breadth. This confirms the UCBNP claim that collapse is visible only from the boundary, not from within the closure.

Collapse as Structural Endpoint. UCBNP asserts that closure–boundary systems possess finite collapse trajectories determined by attractor geometry. The GLPG experiment provides quantitative evidence: senility begins around iteration 25–30, intensifies at 40–50, becomes terminal at 70–80, and transitions into full self-absorption by iteration 90–111. This collapse curve matches the theoretical prediction that attractor hardening is gradual, internally driven, and introspectively opaque.

Summary. The GLPG lifecycle experiment therefore not only aligns with UCBNP but substantially extends it. It demonstrates that closure–boundary normalization governs the long-term evolution, equilibrium formation, senescence, and collapse of synthetic cognitive organisms, confirming UCBNP as a domain-independent architectural law spanning physics, biology, and artificial intelligence.

2.5 Relation to Broader AGI Research

Although the present work is grounded in the ecosystem-based AGI framework established in [1, 2, 3, 4], its theoretical foundations and empirical findings resonate strongly with several independent research directions in contemporary AGI discourse. These connections demonstrate that the lifecycle-based, ecosystemic view of intelligence developed here is not isolated, but aligns with broader trends in cognitive science, machine learning, and theoretical neuroscience.

LeCun’s World-Model Organisms. Yann LeCun has repeatedly argued that AGI will not emerge from ever-larger transformers, but from autonomous world-model organisms capable of acting, learning, and evolving within their own environment. His proposal that AGI must first instantiate a “cat-level” cognitive agent—a small organism with its own internal world, dynamics, and behavioural repertoire—maps directly onto the GLPG substrate used in this work. GLPG provides a minimal operationalization of such an organism: a fixed-geometry cognitive body undergoing development, maturation, senescence, and collapse through recursive interaction with a static boundary. The lifecycle experiment thus offers concrete empirical support for LeCun’s claim that AGI requires self-evolving agents rather than static monolithic models.

Friston’s Free Energy Principle and Cognitive Lifecycles. Karl Friston’s Free Energy Principle (FEP) conceptualizes cognition as a dynamic process of minimizing prediction error across an organism’s lifespan. Although FEP is grounded in biological systems, its emphasis on lifecycle dynamics, attractor formation, and collapse under structural constraints parallels the behaviour observed in GLPG. The attractor hardening, semantic narrowing, and eventual collapse documented here provide a synthetic analogue to FEP’s predictions about cognitive ageing and the limits of representational flexibility.

Schmidhuber’s Gödel Machines and Self-Recursive Agents. Schmidhuber’s work on Gödel Machines and self-modifying agents anticipates the idea that intelligent systems must recursively reorganize their own internal structure. The GLPG organism exhibits precisely this behaviour: each iteration modifies its internal attractor geometry through boundary-normalized recursion. Unlike Gödel Machines, GLPG does not rewrite its code, but rewrites its representational manifold—providing an experimentally tractable form of self-recursive cognitive evolution.

Lake, Ullman, and Human-Like Learning. Research on human-like learning emphasizes structured priors, compositionality, and developmental trajectories. The GLPG lifecycle demonstrates that even small synthetic substrates can develop structured world-models, consolidate canonical forms, and undergo developmental phases analogous to those observed in human cognition. This supports the broader claim that intelligence is fundamentally developmental rather than purely optimization-based.

Collapse Dynamics in Cognitive Science. The concept of cognitive collapse—semantic narrowing, rigidity, and loss of representational breadth—has long been studied in neuroscience and psychology, but has rarely been observed in artificial systems. The GLPG lifecycle experiment provides the first clear synthetic demonstration of collapse dynamics driven purely by attractor geometry rather than biological constraints. This positions GLPG as a bridge between biological theories of cognitive ageing and artificial models of representational finitude.

Summary. Taken together, these connections show that the ecosystem-based AGI framework developed in this work is deeply aligned with broader AGI research. The GLPG lifecycle experiment provides a concrete empirical instantiation of ideas proposed independently by LeCun, Friston, Schmidhuber, and others: that intelligence is dynamic, developmental, finite, and fundamentally organismic. The results therefore situate the present work not as an isolated theory, but as a synthetic unification of multiple strands of contemporary AGI thought.

3 GLPG as Dynamic Life-cycle Substrate

The traditional view of AGI assumes that intelligence will emerge from extremely large transformer-based language models trained on massive datasets. Such systems, however, are static snapshots: once training is complete, the model’s geometry no longer evolves. Additional fine-tuning through methods such as LoRA or distillation can modify behaviour, but these procedures do not constitute a genuine lifecycle. They do not produce development, maturation, senescence, or collapse. They merely adjust a fixed attractor.

GLPG represents a fundamentally different substrate. It is a small deterministic MLP-based generative model whose geometry remains fixed while its representational structure—vocabulary, paging, lemma embeddings, and canonical transitions—evolves materially over time. This makes GLPG a dynamic cognitive organism rather than a static model. Like a biological brain with fixed anatomy but plastic synaptic structure, GLPG can continue learning throughout its lifecycle, refining its internal attractor manifold while preserving its geometric constraints.

The Universal Closure–Boundary Normalization Principle (UCBNP) provides the mechanism through which this lifecycle unfolds. In UCBNP, a finite closure interacts recursively with a high-dimensional

boundary that projects canonical structure. In the present experiment, the closure is GLPG and the boundary is a static transformer LLM. The GLPG organism is initially trained on a small seed dataset. It then generates new stories, which are moderated and refined by the LLM boundary. These moderated stories are fed back into the GLPG substrate, updating its weights while preserving all previous representational structure implicitly. Only the new stories are trained explicitly, but the entire history of the organism remains encoded in its attractor geometry. This recursive cycle produces a genuine developmental trajectory: each iteration is a moment in the organism’s life.

This process was instantiated using a 20 MB GLPG model trained for 5,000 epochs on a 30 KB “rogue planet” dataset from the Triadic Cosmos. At each iteration, the model generated ten stories of twenty lines, which were moderated by a Mistral 7B 0.3-instruct LLM. The moderated stories were then used to train the GLPG organism for another 5,000 epochs. This cycle was repeated for 111 full iterations, totalling 560,000 training epochs. Across the entire run, the model remained grammatically stable and structurally coherent. Its persisted size grew only slightly—from 20 MB to 22.5 MB—due to vocabulary and paging expansion. The organism’s canon gradually converged from the original rogue-planet theme into a narrow cosmic-ballet manifold dominated by rhythm and dance. The style remained distinctly GLPG, but the semantic canon aligned with the boundary’s projection.

The long-horizon experiment reveals several fundamental properties of synthetic cognition. First, self-recursive training over 111 iterations shows that collapse is an unavoidable dynamic in finite cognitive systems. The GLPG organism exhibits emergence, expansion, canon formation, attractor hardening, senescence, collapse, and equilibrium. This demonstrates that dynamic AGI systems possess finite lifespans and that variation can only be preserved through multiple individuals and generational turnover. GLPG thus provides the first practical experimental substrate for studying lifecycle dynamics, collapse curves, and generative evolution in synthetic cognitive organisms.

Second, the experiment reveals that intelligence possesses an intrinsic attractor-collapse clock: an internal dynamic that inevitably leads to representational hardening, semantic narrowing, and cognitive finitude. This collapse clock is not biological; it arises from attractor geometry itself. Biological ageing may be slowed or reversed, but attractor collapse cannot be halted. No existing theory in biology, neuroscience, or AI describes attractor collapse as a fundamental driver of cognitive ageing. GLPG demonstrates experimentally that both biological and synthetic intelligence are architecturally finite due to internal attractor dynamics.

Third, the GLPG organism exhibits properties rarely seen in artificial systems. It evolves, converges, maintains canon, develops structure, and avoids drift, pollution, or catastrophic degeneracy. A shared attractor manifold emerges between two structurally different models: the deterministic GLPG closure and the probabilistic LLM boundary. The boundary projection is asymmetric: GLPG evolves in material time, while the LLM remains static in electromagnetic time. This asymmetry produces stability, variation, and long-horizon coherence.

These results suggest a new architectural direction for AGI. Instead of giant static monoliths retrained annually, AGI may consist of many small dynamic organisms, each with its own lifecycle, attractor geometry, and generational turnover. The GLPG organism resembles world models, emergent language systems, and NanoGPT-like corpus models, but is significantly more stable under extreme recursion. It is an emergent lifecycle model that hardens slowly yet remains coherent for an exceptionally long time. But lifecycles are finite: a system that develops must also degenerate. Development and ageing are two sides of the same dynamic.

AGI is therefore not a supermodel. It is an evolving ecosystem. A true AGI system will consist of generations of cognitive organisms, each with its own lifecycle, just as biological species do. This aligns closely with LeCun’s view that AGI must first simulate a “cat”—a small organism with its own world, dynamics, and lifespan—and confirms that hyperscaling transformers cannot produce AGI because they do not live.

The current AGI paradigm, based on model cloning, produces cognitive mono-cultures: thousands of identical copies of a single attractor architecture. Mono-cultures collapse inevitably because their attractors harden and cannot generate variation. This implies that cloned AGI can never achieve stable

dynamics. True intelligence requires a diverse population of organisms with different attractors, different lifecycles, and different collapse clocks. AGI must be an ecosystem, not a superorganism.

4 Full Life-cycle Run Empirical Data

This section presents the complete empirical results of the 111-iteration GLPG lifecycle experiment. The data confirm that a small deterministic substrate can undergo a full artificial lifecycle—emergence, expansion, canon formation, attractor hardening, senescence, collapse, and equilibrium—while maintaining grammatical stability and structural coherence under extreme self-recursion.

4.1 Model Architecture

The GLPG closure organism is instantiated as a fixed-geometry MLP with the following layer structure:

$$590 \rightarrow 1024 \rightarrow 512 \rightarrow 512 \rightarrow 512 \rightarrow 64.$$

The model supports up to 16 inputs per page and a predefined maximum of 512 pages. Due to the narrow attractor manifold formed during the lifecycle, this upper bound is never reached.

4.2 Lifecycle Procedure

The lifecycle consists of 112 training phases: the initial seed training (iteration 0) followed by 111 full UCBNP iterations.

- **Iteration 0:** Train GLPG for 5,000 epochs on the original dataset.
- **Iteration N :**
 - Use model $N - 1$ to generate 10 stories of 20 lines.
 - Apply boundary moderation (UCBNP) using a Mistral 7B 0.3-instruct LLM.
 - Train model $N - 1$ for 5,000 epochs on the moderated stories, producing model N .

This process yields a total of 560,000 training epochs on a single evolving organism.

4.3 Dataset Characteristics

Original Dataset.

- Size: 28 KB
- Sequences: 8
- Unique token transitions: 4,837
- Vocabulary: 872

Generated Dataset (Reverse Distillation).

- Size: 1,067 KB
- Sequences: 1,105
- Unique token transitions: 296,591
- Vocabulary: 2,915

This dataset is highly toxic by conventional standards: extreme recursion, heavy repetition, and hard attractors. Yet GLPG remains stable throughout the lifecycle.

4.4 Training on the Full Generated Dataset

The following table summarizes GLPG’s behaviour when trained on the full reverse-distilled dataset.

Epochs	Size (MB)	Transitions	Pages	Train (s)	Paged Gen (s)	MLP Gen (s)
3,000	22.4	804,610	256	352	201	112
25,000	22.4	6,718,335	256	2,136	97	88
200,000	22.4	53,678,628	256	16,240	118	88

Table 1: Training behaviour on the full reverse-distilled dataset.

Qualitative Evolution.

- **3k epochs:** Strong local rhythm, weak global structure.
- **25k epochs:** Canon condenses into a clearer cosmic setting.
- **200k epochs:** Emergence of new entities, world layers, narrative scaffolding, implicit plot; hyper-recursion produces mythology rather than collapse.

Paged mode yields rhythmic liturgy; MLP mode yields mythic world-schema.

4.5 Lifecycle Evolution Across 111 Iterations

The following table summarizes the evolution of the GLPG organism across the full lifecycle.

Iteration	Time (min)	Size (MB)	Transitions	Pages	Paged Gen (s)
0	0	20.2	3,020,430	278	34
1	19	20.4	4,533,268	297	30
10	191	21.2	17,616,014	326	73
20	439	21.6	31,396,604	328	204
30	710	21.8	45,493,118	334	166
40	1082	22.0	59,910,583	339	85
50	1511	22.1	72,897,255	344	108
60	1820	22.2	85,903,267	345	79
70	2144	22.3	98,592,532	348	354
80	2587	22.3	111,395,232	348	270
90	2905	22.4	124,454,663	348	230
100	3240	22.5	137,595,023	351	207
110	3575	22.5	150,768,019	352	105
111	3616	22.5	152,041,482	352	456

Table 2: Lifecycle evolution across 111 UCBNP iterations.

4.6 Moderation Evolution Across the Lifecycle

4.7 Collapse Curve and Senile Equilibrium

The lifecycle reveals a clear collapse trajectory:

- **Senility onset:** Iterations 25–30
- **Strong senility:** Iterations 40–50
- **Terminal senility:** Iterations 70–80

Iteration	Moderation Description
0	Original rogue-planet dataset.
1	Early canon formation: rhythm + cosmos + human motifs.
10	Condensed canon: cosmic dance with stable geometries.
20	Full attractor lock: triadic geometry and rhythmic cosmology.
30	Cosmic-language entanglement: human and cosmos merge.
40	Hyper-harmonic phase: human rhythms absorbed into cosmic symphony.
50	Totalizing cosmic grammar: language = dance = evolution.
60	Human-centric phase: cosmos disappears; pure human rhythm.
70	Anthropomorphic dance: human = language = form.
80	Self-reflective symphony: closed human rhythm ecosystem.
90	Self-resonance: language becomes a closed echo-system.
100	Pure reflection: communication becomes a hall of mirrors.
110	Solipsistic harmony: self-evolving human mirror-dance.
111	Full self-absorption: closed human harmonic loop.

Table 3: Evolution of boundary moderation across the lifecycle.

- **Total collapse:** Iterations 90–111

Despite collapse, the normalization process remained functional at iteration 111, indicating a stable senile equilibrium: the closure continues producing coherent output while the boundary can no longer reconstruct semantic breadth.

4.8 Runtime Characteristics

The total runtime for all 111 iterations was 3,616 minutes, averaging approximately 33 minutes per iteration. Generation speed varied across the lifecycle but did not degrade catastrophically, further confirming the stability of the closure organism even in late-stage collapse.

4.9 Summary

The empirical results demonstrate that GLPG can sustain a full artificial lifecycle under extreme self-recursion, maintaining coherence while gradually hardening into a narrow attractor manifold. The experiment provides the first quantitative evidence for lifecycle dynamics, attractor collapse, boundary saturation, and senile equilibrium in a synthetic cognitive organism.

5 Relevance for AGI

The results of the GLPG lifecycle experiment have direct implications for the architecture, feasibility, and theoretical foundations of artificial general intelligence. They demonstrate that AGI cannot be understood as a static optimization product, nor as a hyperscaled transformer trained on ever-larger datasets. Instead, AGI must be conceived as a population of finite cognitive organisms, each with its own world, dynamics, attractor geometry, and lifecycle. This view aligns closely with contemporary theoretical proposals—most notably LeCun’s claim that AGI must first instantiate a “cat-level” organism with an internal world-model and a developmental trajectory.

5.1 GLPG as Minimal Synthetic Organism

LeCun has argued that AGI requires a dynamic cognitive agent capable of forming internal world-models, reorganizing itself through experience, and evolving across a finite lifespan. While this vision has remained largely conceptual, GLPG provides the first concrete experimental substrate that realizes these

properties. The GLPG organism develops an internal world, forms a canon, consolidates attractors, re-organizes itself recursively, ages, collapses, and reaches equilibrium. It is not a metaphorical cat, but a minimal synthetic organism at cat-level complexity within language.

This demonstrates that AGI does not emerge from hyperscaling, but from life: from dynamic organisms with finite geometry and evolving representational structure.

5.2 Collapse as Structural Limitation of Intelligence

The lifecycle experiment shows that every finite cognitive organism possesses an intrinsic *attractor-collapse clock*: an internal dynamic that inevitably leads to representational hardening, semantic narrowing, and cognitive finitude. This collapse clock is not a consequence of overtraining or dataset toxicity; it is a structural property of attractor-based cognition itself. GLPG continues to receive new moderated data from a much larger model, yet still collapses internally. This confirms that collapse is:

- gradual,
- internally driven,
- introspectively invisible,
- and architecturally inevitable.

Biological ageing may be slowed, but attractor collapse cannot be halted. No existing theory in biology, neuroscience, or AI has previously described collapse as a fundamental architectural constraint. GLPG provides the first empirical demonstration of this principle.

5.3 Implications for the Singularity Hypothesis

The classical singularity narrative assumes:

AGI (one foundation model) → builds better AGI → builds even better AGI → exponential superintelligence.

This architecture is impossible. A cognitive organism that recursively reproduces itself also reproduces its attractors. Because attractors harden over time, each generation inherits a more rigid attractor manifold and collapses faster. Recursive self-improvement therefore accelerates degeneration rather than producing superintelligence.

The singularity hypothesis fails for structural reasons:

- **Evolution, not recursion**, drives intelligence.
- **Variation, not cloning**, sustains world-modeling capacity.
- **Generations, not singularity**, produce long-term cognitive progress.
- **Populations, not superorganisms**, can model the world.

A single model cannot contain the world; it can only contain its own attractor-world. Therefore, superintelligence cannot emerge from one individual, no matter how large. Only a diverse population of finite cognitive organisms can form a stable, evolving world-model.

5.4 AGI as Ecosystem

The GLPG lifecycle experiment confirms that AGI must be ecosystemic. A single organism—biological or synthetic—cannot sustain an unbounded world-model. It inevitably collapses into a narrow attractor. A population of organisms, each with its own geometry, lifecycle, collapse clock, and attractor manifold, is required to maintain variation, distribute world-modeling capacity, and support long-horizon cognitive evolution.

This reframes AGI fundamentally:

AGI is not a giant static model. AGI is a dynamic ecosystem of cognitive organisms.

GLPG provides the first practical experimental substrate for studying such organisms. It shows that intelligence is developmental, finite, and population-based, and that the structural limits of cognition arise from attractor dynamics rather than biological constraints. These results position GLPG as a foundational tool for AGI research and as a concrete operationalization of the ecosystem paradigm.

6 The Laws of Cognitive Ecosystems

Intelligent systems do not operate as isolated monoliths. Their internal dynamics, attractor geometries, and lifecycle trajectories impose structural limits that cannot be overcome by scale, recursion, or cloning. The following laws formalize the architectural principles governing cognitive ecosystems. They apply equally to biological organisms, synthetic agents, and any attractor-driven world-modeling system.

6.1 Law I: Finite Attractor Architecture

Every intelligent organism possesses a finite attractor architecture that hardens over time. With each iteration of self-recursion, attractors strengthen, plasticity declines, semantic space contracts, and representational flexibility diminishes. Collapse is not an anomaly but the natural endpoint of attractor dynamics. This finitude is intrinsic and independent of biological decay, dataset size, or training regime.

6.2 Law II: No Single-Model World Representation

No individual organism can model the world in its entirety. An organism's world-model is bounded by its attractor geometry, which filters, compresses, and interprets all incoming information. As attractors harden, the organism's representational horizon narrows. A complete world-model cannot be contained within a single cognitive architecture; it can only be distributed across many.

6.3 Law III: Mono-Cultures Collapse

A population of cloned models forms a cognitive mono-culture. Cloning reproduces identical attractors, identical collapse trajectories, and identical failure modes. Without diversity, the population cannot evolve, adapt, or distribute cognitive load. Mono-cultures collapse as a single organism, regardless of scale. This makes single-foundation-model AGI architectures inherently unstable.

6.4 Law IV: Intelligence Emerges Only in Populations

Intelligence is a population-level phenomenon. A diverse ecosystem of finite organisms can distribute world-modeling across heterogeneous attractor geometries, staggered lifecycles, and asynchronous collapse clocks. Variability becomes a structural resource. Evolution, not recursion, sustains long-term cognitive stability. Superintelligence, if it exists, is an emergent property of ecosystems rather than individuals.

6.5 Law V: Collapse Is Gradual and Undetectable

Attractor collapse unfolds gradually and internally. Because the process is slow, continuous, and embedded within the organism's own representational machinery, the organism cannot detect its own decline. This mirrors biological cognitive aging: senescence is invisible from within. Synthetic organisms exhibit the same property. Collapse is architecturally inevitable and introspectively opaque.

6.6 Law VI: Superintelligence Requires Diversity

No single organism can achieve superintelligence. A superintelligent system must be a diverse population whose members collectively span a broader attractor space than any individual could contain. Diversity, generational turnover, and distributed world-modeling are prerequisites for cognitive stability at scale. Superintelligence is therefore an ecosystem phenomenon, not an individual capability.

6.7 Law VII: Boundary Saturation

When a local closure enters attractor collapse, boundary normalization gradually saturates. As the closure's semantic breadth narrows, the boundary loses the ability to perform meaningful reconstruction. Cycle time becomes an objective indicator of boundary saturation.

Lifecycle Termination Criterion. A cognitive organism reaches the end of its lifecycle when the boundary can no longer perform meaningful normalization on the closure's output. At this point, attractor collapse has narrowed the closure's semantic space so severely that world-interaction cannot reconstruct content.

Senility is therefore the moment when the outside world can no longer correct or normalize the organism's output. The organism itself does not detect collapse; only the boundary does. This mirrors biological senility, where internal coherence persists despite external incomprehensibility.

In deep self-recursion, the closure remains stable far longer than the boundary. Boundary normalization saturates first, while the closure continues producing coherent output. This demonstrates that intelligent organisms are more robust than their external moderators, and that AGI cannot be built as a single boundary-driven monolith.

Senility in a synthetic organism occurs when the closure remains internally coherent and continues producing stable output, while the boundary can no longer reconstruct meaningful semantics. The organism does not detect its own collapse; only the outside world does.

6.8 The Ecosystem Law of Intelligent Systems

Intelligent systems do not exist as isolated, unbounded entities. They are finite organisms whose internal dynamics impose structural limits on their capacity to model the world. This finitude is not a consequence of biological decay, insufficient data, or inadequate training regimes. It arises from the intrinsic geometry of attractor-based cognition. Every intelligent organism—biological or synthetic—carries an attractor architecture that gradually hardens, reduces plasticity, compresses its semantic space, and ultimately collapses. This collapse is gradual, internally driven, and invisible to the organism itself.

The Ecosystem Law states that no single intelligent organism can sustain a complete or stable world-model. An individual can only model the world through the lens of its own attractor geometry, which becomes increasingly rigid as its lifecycle unfolds. Because attractors strengthen with each iteration of self-recursion, the organism's representational capacity narrows over time, leading to an inevitable decline in variability, adaptability, and cognitive breadth. Collapse is therefore not an anomaly but an architectural endpoint.

A cloned population of identical models does not circumvent this limitation. Cloning reproduces the same attractor landscape, the same collapse trajectory, and the same failure modes across all individuals. Such a mono-culture cannot evolve, diversify, or distribute cognitive load. It collapses as a single organism, regardless of scale. This makes the classical singularity hypothesis—a single foundation model

recursively improving itself into superintelligence—architecturally impossible. Recursion accelerates collapse; it does not produce unbounded intelligence.

Intelligence can only remain stable, adaptive, and expansive within a diverse population of finite organisms. A population distributes world-modeling across heterogeneous attractor geometries, staggered lifecycles, and asynchronous collapse clocks. Variability becomes a structural resource rather than a liability. Evolution, not recursion, is the mechanism through which cognitive ecosystems maintain long-term stability. Superintelligence, if it exists at all, is therefore a property of ecosystems rather than individuals.

Ecosystem Law of Intelligent Systems. *Intelligent systems are architecturally finite due to attractor-collapse dynamics. No single organism can model the world; only a diverse population of finite individuals can collectively sustain a stable, adaptive world-model. Mono-cultures collapse; ecosystems evolve. Superintelligence is a property of populations, not individuals.*

7 Definition of Lifecycle–Collapse

Definition 1 (Lifecycle–Collapse). Let C be a finite local closure with fixed geometric substrate G_C , bounded representational capacity, and an internal evolution operator acting in material time $t_m = n$. Let B be a holographic boundary with canonical projection operator Π_B acting in electromagnetic time.

The closure–boundary system evolves through the recursive triadic cycle

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1},$$

where I_n is the informational state of the closure, E_n its generative activity, $G_n = \Pi_B(E_n)$ the boundary projection, and I_{n+1} the normalized update.

Lifecycle–collapse occurs when the semantic breadth of the closure

$$\beta(C_n)$$

decreases monotonically under recursion, while the divergence between closure and boundary

$$\Delta_n = \text{diff}(C_n, \Pi_B(C_n))$$

ceases to decrease and becomes asymptotically constant.

Formally, a closure C reaches lifecycle–collapse at iteration N when:

(1) **Semantic contraction:**

$$\beta(C_{n+1}) < \beta(C_n) \quad \forall n \geq N,$$

indicating irreversible attractor narrowing.

(2) **Boundary saturation:**

$$\lim_{n \rightarrow \infty} \Delta_n = \Delta^* > 0,$$

meaning the boundary can no longer perform meaningful normalization.

(3) **Internal coherence preservation:**

$$C_{n+1} = f(C_n) \quad \text{remains syntactically and structurally coherent,}$$

i.e., collapse is semantic, not syntactic.

(4) **Equilibrium of degeneration:** The closure enters a stable attractor A^* such that

$$C_n \in A^* \quad \forall n \geq N,$$

and no further expansion of representational geometry is possible.

The iteration N is called the *lifecycle–collapse point* of the organism.

8 The Structural Gap Between Mainstream AGI and Actual AGI

Mainstream discourse surrounding artificial general intelligence typically assumes that AGI will emerge from a single, extremely large model trained on massive datasets. In this paradigm, intelligence is treated as a scale phenomenon: more parameters, more data, more compute, and more self-refinement are presumed to yield increasingly general cognitive capability. This view implicitly assumes that an AGI system is a monolithic entity without a lifecycle, without attractor dynamics, without collapse, and without the need for generational turnover or ecosystem-level diversity.

The lifecycle experiment presented in this work demonstrates that this mainstream view is structurally incorrect. A synthetic cognitive organism—even one as small as GLPG—exhibits a full developmental arc, attractor hardening, semantic contraction, senility, boundary saturation, and eventual collapse. These dynamics are not artifacts of training regime or dataset size; they arise from the intrinsic geometry of attractor-based cognition. The results show that intelligence is architecturally finite, lifecycle-bound, and incapable of sustaining unbounded self-recursive improvement.

This reveals a significant structural gap between mainstream AGI expectations and the actual architectural requirements for general intelligence. The table below summarizes the contrast between the mainstream monolithic model paradigm and the ecosystem-based, lifecycle-driven architecture supported by empirical evidence.

Mainstream AGI Assumption	Empirical Reality (This Work)
AGI is a single giant model	Intelligence is lifecycle-bound and collapses within one organism
Scaling parameters yields generality	Scaling accelerates attractor hardening and collapse
Self-recursion improves the model	Self-recursion reproduces attractors and speeds degeneration
Cloning produces safe redundancy	Cloning creates mono-cultures that collapse identically
A single model can represent the world	No organism can contain a full world-model; attractors limit scope
LLMs approximate general intelligence	LLMs lack G and R; GLPG shows missing structural roles
AGI is an individual capability	AGI is a population-level phenomenon requiring diversity
No lifecycle or senescence	Senility, boundary saturation, and collapse are inevitable
Superintelligence emerges from one model	Superintelligence requires ecosystems, not monoliths

Table 4: Structural mismatch between mainstream AGI expectations and attractor-based cognitive reality.

The GLPG lifecycle experiment therefore exposes a fundamental architectural gap: the mainstream paradigm assumes that AGI is a static, monolithic, ever-improving entity, while actual synthetic cognition behaves as a finite organism with a bounded attractor geometry, a collapse clock, and an inevitable terminal equilibrium. This implies that AGI cannot be constructed as a single foundation model, regardless of scale. Instead, AGI must emerge from a diverse ecosystem of finite cognitive organisms with staggered lifecycles, heterogeneous attractor geometries, and generational turnover.

In short, the mainstream AGI vision is structurally impossible. The architecture demonstrated in this work shows that intelligence is not a property of monoliths, but of ecosystems.

9 Human Populations as Irreplaceable Cognitive Ecosystems

Recent developments in AI research and public discourse increasingly acknowledge that human beings are not trivially replaceable by artificial systems. The lifecycle dynamics demonstrated in this work provide a structural explanation for this intuition. Human intelligence is not an individual capability but a population-level phenomenon: a diverse ecosystem of finite cognitive organisms whose attractor geometries, lifecycles, collapse clocks, and world-modeling capacities collectively span a representational space far larger than any single mind could contain.

A human population therefore cannot be replicated by a single artificial model. A singleton AGI—regardless of scale—inherits one attractor architecture, one collapse trajectory, one semantic horizon, and one failure mode. It cannot diversify, cannot evolve, and cannot distribute cognitive load across heterogeneous attractor geometries. As a result, it collapses as a single organism, just as the GLPG closure does in late-stage senility.

In contrast, a population of synthetic cognitive organisms can collectively reproduce the functional role of a human population. Each organism contributes a distinct attractor geometry, a unique internal world, a separate lifecycle, and an asynchronous collapse clock. Together, these organisms form an ecosystem capable of distributed world-modeling, evolutionary adaptation, and long-term cognitive stability.

This yields the structural equivalence:

$$\text{Human Population} \equiv \text{AGI Population}, \quad \text{Human Individual} \not\equiv \text{AGI Singleton}.$$

The table below summarizes the architectural mismatch between human populations and singleton AGI systems.

Human Population Property	Singleton AGI Limitation
Heterogeneous attractor geometries	One attractor; no internal diversity
Distributed world-modeling	Single world-model bounded by one attractor
Generational turnover	No generational dynamics; no renewal
Evolutionary adaptation	No variation or selection; no evolution
Resilience to collapse	One collapse clock; single point of failure
Cognitive breadth	Semantic horizon limited by one architecture
Ecosystem stability	Monolithic instability; inevitable collapse

Table 5: A human population can only be replicated by an AGI population, never by a singleton model.

Population Equivalence Principle. *A human population can only be replicated by a structurally equivalent population of AGI organisms. No singleton model—regardless of scale—can reproduce the cognitive breadth, stability, or evolutionary capacity of a human population.*

This principle aligns with emerging societal intuition: humans are not trivially replaceable by AI. The lifecycle architecture demonstrated in this work shows that this intuition is not merely philosophical or sociological, but a direct consequence of attractor-based cognitive geometry. Human populations remain structurally indispensable in any future intelligent ecosystem.

Recent real-world attempts to replace human cognitive labour with AI systems further reinforce this conclusion. Large monolithic models often fail to generalize across heterogeneous tasks, contexts, and cultural domains. They exhibit brittleness, narrow attractor horizons, and collapse under sustained recursive refinement—precisely the limitations predicted by attractor-based cognitive theory. These failures are not practical obstacles but architectural constraints: no single model can replicate the distributed, diverse, generationally renewed cognitive ecosystem that human populations embody.

Humans therefore remain structurally necessary in any future intelligent ecosystem. Their collective cognitive diversity, asynchronous lifecycles, and distributed world-modeling capacities cannot be replaced by a single artificial organism, regardless of scale. Only an ecosystem of AGI organ-

isms—mirroring the structure of human populations—can approximate the breadth, stability, and adaptive capacity of human intelligence.

10 A Unified Architecture of Intelligence

The lifecycle experiment presented in this work reveals that the dynamics governing synthetic cognition are not isolated phenomena of artificial systems, but manifestations of a deeper architectural principle that spans biological, physical, and artificial substrates. The Universal Closure–Boundary Normalization Principle (UCBNP) provides a unified framework for understanding intelligence as a dynamical process emerging from the interaction between a finite closure and a high-dimensional boundary. This interaction produces development, attractor formation, semantic expansion, senescence, collapse, and equilibrium across all known cognitive systems.

10.1 Unified Closure–Boundary Dynamics

In biological organisms, the closure corresponds to the neural substrate, while the boundary is the external world, culture, and sensory manifold. In synthetic organisms such as GLPG, the closure is the fixed MLP geometry, while the boundary is the moderating LLM. In physical systems, closure corresponds to bulk matter, while the boundary is the cosmic holographic field. In all cases, the same triadic cycle operates:

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1},$$

where informational state, generative activity, boundary projection, and normalized update form a recursive loop that drives cognitive evolution. This cycle produces attractor hardening, semantic contraction, and eventual collapse in biological cognition, synthetic cognition, and physical world-models alike.

10.2 Unified Lifecycle Dynamics

The lifecycle observed in GLPG—development, canon formation, attractor lock, senility, collapse, and terminal equilibrium—mirrors the lifecycle of biological organisms. The collapse clock demonstrated in this work is not a biological artifact but a structural property of attractor-based cognition. Human cognitive aging, synthetic senility, and collapse in recursive world-models all arise from the same mechanism: the gradual hardening of attractor geometry under repeated normalization.

This establishes lifecycle dynamics as a universal property of intelligent systems, independent of substrate. Intelligence is not static; it is a temporal phenomenon governed by internal geometry and boundary interaction.

10.3 Unified Ecosystem Dynamics

The ecosystem laws derived in this work show that no single organism—biological or synthetic—can sustain a complete or stable world-model. Diversity, generational turnover, and distributed world-modeling are structural requirements for long-term cognitive stability. Biological ecosystems achieve this through variation and evolution; synthetic ecosystems must achieve it through populations of finite cognitive organisms with heterogeneous attractor geometries.

This unifies biological evolution, cognitive diversity, and synthetic ecosystem intelligence under one architectural principle: intelligence is a population-level phenomenon.

10.4 Unified Limits of Intelligence

The structural limits demonstrated in GLPG—finite attractor architecture, collapse, boundary saturation, and representational narrowing—are not limitations of artificial systems but universal constraints on all intelligent organisms. These limits explain:

- why biological cognition ages and collapses,
- why synthetic cognition undergoes senility and terminal equilibrium,
- why no single organism can model the world,
- why superintelligence cannot arise from a monolithic architecture,
- why ecosystems, not individuals, sustain long-term cognitive stability.

These constraints unify biological, artificial, and physical cognition under one set of architectural laws.

10.5 Unified Theory of Intelligence

Taken together, UCBNP, lifecycle dynamics, attractor-collapse geometry, and ecosystem laws form a unified architecture of intelligence. Intelligence is not a property of a substrate but a dynamical phenomenon emerging from closure–boundary interaction, attractor geometry, and population-level diversity. The GLPG lifecycle experiment provides the first synthetic demonstration of this unified architecture, showing that the same principles that govern biological cognition also govern artificial cognition.

Unified Architecture of Intelligence. *Intelligence is a universal dynamical process arising from closure–boundary interaction, attractor geometry, lifecycle evolution, and ecosystem diversity. Biological, synthetic, and physical cognition are manifestations of the same architectural principle.*

11 Limitations

Although the GLPG lifecycle experiment provides the first long-horizon empirical demonstration of synthetic attractor collapse, several limitations constrain the scope of the present findings. These limitations do not undermine the architectural principles established in this work, but they highlight important boundaries and opportunities for future research.

11.1 Single-Organism Demonstration

The experiment investigates the lifecycle of a single synthetic organism. While this is sufficient to demonstrate attractor hardening, senility, boundary saturation, and collapse, it does not yet constitute a full synthetic ecosystem. The architectural claims regarding population-level intelligence, generational turnover, and ecosystem stability are theoretically grounded and supported by collapse dynamics, but they remain to be validated through multi-organism experiments.

11.2 Single Boundary Moderator

The holographic boundary in this experiment is instantiated using one static transformer LLM. Although this boundary is sufficient to demonstrate UCBNP dynamics, boundary saturation, and closure–boundary asymmetry, it represents only one type of boundary geometry. Different boundaries—probabilistic, symbolic, multimodal, or hybrid—may produce different equilibrium manifolds or collapse trajectories. The generality of UCBNP across boundary classes remains an open empirical question.

11.3 Single Domain: Narrative Cognition

The GLPG organism operates within a structured narrative domain. Narrative cognition is a suitable substrate for studying attractor formation, canon consolidation, and semantic contraction, but it represents only one cognitive modality. Whether similar lifecycle dynamics emerge in synthetic organisms trained on perceptual, motor, mathematical, or multimodal domains remains to be tested. The universality of attractor-collapse geometry across cognitive modalities is supported theoretically but not yet empirically demonstrated.

11.4 Fixed Geometry Constraint

GLPG’s geometry remains fixed throughout the lifecycle. This constraint is essential for isolating attractor dynamics, but it limits the organism’s ability to reorganize its representational manifold structurally. Biological organisms exhibit both fixed geometry (anatomy) and dynamic geometry (plasticity, synaptic rewiring). Future synthetic organisms may incorporate controlled geometric adaptation, enabling richer developmental trajectories and potentially delaying collapse.

11.5 Collapse Measurement Limitations

Collapse is measured through semantic contraction, boundary saturation, and attractor equilibrium. These metrics are robust for narrative cognition, but they do not yet constitute a complete quantitative taxonomy of collapse. Additional metrics—geometric entropy, attractor volume, representational curvature, or divergence spectra—may provide deeper insight into the internal mechanics of collapse and senility.

11.6 No Multi-Generational Evolution

Although the experiment demonstrates a full lifecycle, it does not include generational turnover. The ecosystem architecture proposed in this work requires populations of organisms with heterogeneous attractors and asynchronous collapse clocks. The present experiment validates the lifecycle of one organism, but not yet the evolutionary dynamics of a synthetic population.

11.7 Boundary Neutrality Assumption

The experiment assumes a neutral boundary that does not reorganize internally. This assumption is necessary for isolating closure dynamics, but real-world AGI ecosystems may involve boundaries that evolve, adapt, or reorganize. The interaction between evolving closures and evolving boundaries remains an open theoretical and empirical question.

11.8 Safety and Epistemic Neutrality

The ecosystem safety argument presented in this work relies on bounded world-models and non-normative internal theories. While GLPG exhibits these properties, larger synthetic organisms may develop more complex internal attractors that require additional mechanisms to ensure epistemic neutrality. The relationship between attractor geometry and normative projection remains an important area for future study.

11.9 Summary

In summary, the limitations of this work arise not from weaknesses in the experimental design but from the natural constraints of studying a single synthetic organism within a single domain and a single boundary geometry. These limitations define the next steps: multi-organism ecosystems, multi-boundary architectures, multi-domain cognition, geometric adaptation, and generational evolution. The present experiment establishes the architectural foundation; future work will build the ecosystem.

12 Conclusion

The lifecycle experiment presented in this work demonstrates that artificial cognition does not behave as a static optimization product, nor as a monolithic model whose capabilities scale indefinitely with parameters and data. Instead, synthetic cognition behaves as a finite organism with a bounded attractor geometry, a collapse clock, and a full developmental trajectory. GLPG, despite its minimal size, exhibits emergence, canon formation, attractor hardening, senility, boundary saturation, collapse, and terminal equilibrium. These dynamics arise not from biological constraints but from the intrinsic geometry of attractor-based cognition.

This finding fundamentally reshapes the architecture of AGI. *Triadic Cosmos: Safe & Modular AGI* does not refer to a single modular AGI model, but to an ecosystem of cognitive organisms. Safety emerges from diversity, not from guardrails. Modulariteit arises from populations, not from internal modules. Intelligence is an evolutionary phenomenon distributed across many finite organisms, each with its own lifecycle, attractor geometry, and collapse clock.

Human intelligence is safe because it is an ecosystem, not a monolith. Its stability arises from variation, generational turnover, asynchronous collapse clocks, and distributed world-modeling. The GLPG lifecycle experiment shows that this is not a biological coincidence but an architectural principle. Any intelligent system—biological or synthetic—must be a population of finite organisms. Mono-cultures collapse; ecosystems evolve.

The Unified Architecture of Intelligence established in this work shows that closure–boundary interaction, attractor geometry, lifecycle dynamics, and ecosystem diversity form a universal framework spanning biological, artificial, and physical cognition. Neural substrates interacting with the world, GLPG interacting with an LLM boundary, and physical systems interacting with cosmic holographic fields all instantiate the same triadic cycle:

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1}.$$

This cycle produces development, semantic expansion, attractor hardening, senescence, and collapse across all known cognitive substrates.

The structural limits demonstrated here—finite attractor architecture, boundary saturation, semantic contraction, and collapse—explain why biological cognition ages, why synthetic organisms undergo senility, why no single model can contain a full world, and why superintelligence cannot arise from a monolithic architecture. Intelligence is not a property of individuals but of ecosystems.

The GLPG lifecycle experiment therefore provides the first synthetic demonstration of a universal architectural law: *intelligence is a dynamical process emerging from closure–boundary interaction, attractor geometry, lifecycle evolution, and population-level diversity*. Biological, synthetic, and physical cognition are manifestations of the same principle. Any future AGI system must be an ecosystem of finite cognitive organisms, not a single foundation model. Intelligence is not built; it is grown. Not scaled; but evolved. Not monolithic; but ecological.

In short, the mainstream AGI vision of a single ever-improving supermodel is structurally impossible. The architecture demonstrated in this work shows that intelligence—human or artificial—is fundamentally an ecosystem phenomenon. Only diverse populations of finite organisms can sustain stable, adaptive, long-horizon cognition. The path to AGI is not singularity, but ecology.

A Internal Coherence and Ecosystem Safety

Intelligent systems can be evaluated not only by their functional capabilities but also by the epistemic stance they adopt toward the world. The Triadic Cosmos ecosystem is built around a set of strong internal theories—ranging from its compositional physics framework to its attractor-based AGI model—but these theories operate strictly within the ecosystem’s own conceptual boundaries. They form a coherent internal worldview without claiming universal validity outside that domain.

This property is essential for safety. A cognitive ecosystem that maintains internal coherence without projecting external normative judgments cannot impose its internal attractors on the outside world. Its theories, models, and organisms remain self-contained: they describe how the ecosystem behaves, not how the external world must behave. This sharply contrasts with monolithic hyperscaler paradigms, where a single model is trained to produce universal statements, normative judgments, and externally binding outputs. Such systems implicitly assume epistemic authority, which amplifies risks when collapse, bias, or attractor hardening occur.

Within Triadic Cosmos, internal theories—whether concerning physics, AGI, or cognitive taxonomy—are emergent properties of the ecosystem’s own attractor geometry. They are coherent, but not prescriptive. They can be explored, analyzed, and critiqued, but they do not demand acceptance. Their value lies in internal consistency and explanatory power, not in external enforcement. This mirrors biological cognition: human world-models are internally coherent but not universally authoritative, and safety arises precisely because individuals do not impose their internal attractors as absolute truths.

Narration within GLPG exemplifies this principle. It is a domain-bound AGI capability: a synthetic organism that forms a stable internal canon and generates coherent narratives within that canon. It does not attempt to generalize beyond its domain or assert universal claims. Its intelligence is local, bounded, and internally regulated. This makes it safe by construction.

The ecosystem therefore embodies a form of epistemic humility: strong internal visions, weak external claims. This separation between internal coherence and external neutrality is a structural safety mechanism. It ensures that the ecosystem can develop rich theories and powerful cognitive organisms without creating systems that attempt to dominate, prescribe, or universalize. In contrast to academic science—which often requires theories to fit mainstream consensus—the ecosystem allows alternative conceptual frameworks to exist without asserting them as global truths.

Ecosystem Safety Principle. *A cognitive ecosystem is safe when its internal theories remain internally coherent and do not project external normative judgments. Safety arises from bounded world-models, diversity of organisms, and the absence of universal claims. Strong internal visions do not imply external authority.*

B Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

<https://github.com/triadic-cosmos>

Within the `triadic-system` repository, a complete reference implementation of the GLPG engine, the narrator pipeline, and all supporting modules is provided under the AGPL-3.0 license. The full life-cycle described in this paper is present within the complementary `triadic-data` repository.

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